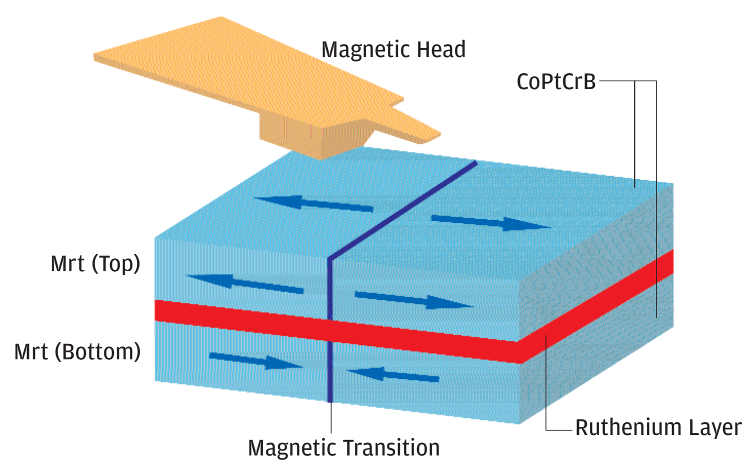
The highest areal density with LR has crossed 100 gigabits psi. And researchers now believe that 120 gigabits psi will be 'too much', and that at that point, PMR will become dominant

arrangement allows for magnetic domains (or bits) to be more closely packed.

So how does it work? First off, recall that the superparamagnetic effect originates from the shrinking volume of the magnetic grains that compose the storage properties of the media. The magnetic grains represent the bits that are stored as alternating magnetic orientations. To increase densities while maintaining acceptable performance, designers have shrunk the media's grain diameters and decreased the thickness of the media. The smaller grain volume makes them increasingly susceptible to thermal fluctuations, which decreases the signal sensed by the drive's head. And if the signal reduction is great enough, data could be lost over time to the superparamagnetic effect.

Now, the precise thickness of the ruthenium in AFC media causes the magnetisation in each of the magnetic layers to be coupled in opposite, or anti-parallel, directions, which constitutes 'antiferromagnetic coupling'. ('Antiferromagnetically-coupled media' is AFC.) Refer to the diagram above.

A Magnetic Head Reading AFC Media



The effective Mrt equals Mrt(Top) - Mrt(Bottom). The ruthenium layer is sandwiched between the two layers. 'CoPtCrB' is cobalt-platinum-chromium-boron. The point here is that the effective Mrt needs to be reduced if the bits are packed more closely together and if the magnetic transitions are still to be read accurately

When reading data as it flies over the rotating disk, the head senses the magnetic 'transitions' in the magnetic media. The strength of this signal is proportional to the media's 'magnetic thickness'—the product of the media's remnant magnetic moment density *Mr* (think of this as simply how strongly the media has been magnetised) and its physical thickness, *t*. As the data density increases, the magnetic thickness-*Mr* x *t*, or *Mrt*—must be decreased so that the closely-packed transitions will be sharp enough to be read clearly. This, in turn, is because the 'transitions' will not be recorded nearly as well if the signal amplitude is too high.

At this point it's better explained with equations. It should be obvious that if *Mrt* is to be reduced, you can either reduce *Mr* or you can reduce *t*. Reducing *Mr* is not an option because that means lower magnetisation, so *t* has to be reduced. And when *t* (the thickness of the media) is reduced too much, as would be done traditionally, the superparamagnetic effect would come in. If it weren't for the ruthenium layer, that is.

With the ruthenium layer in place, it turns out that the effective *Mrt*, or *Mrt(eff)*, is given by *Mrt (eff) = Mrt (top) - Mrt (bottom)*. This is the killer equation: this property of AFC media permits the overall *Mrt* to be reduced, and its data density increased, independent of its overall physical thickness. Thus for a given areal density, the *Mrt* of the top magnetic layer of AFC media can be relatively large compared with single-layer media, permitting larger grain volumes—which, of course, are inherently more thermally stable.

Two additional advantages of AFC media are that it can be made using existing production equipment at little or no additional cost, and that its writing and read-back characteristics are similar to conventional longitudinal media.

Mass production of the first hard drives with AFC media began in May 2001 with Hitachi GST's 2.5-inch Travelstar 15GN and 30GN, which featured 15 GB per platter, and an areal density of 25.7 gigabits psi. And within three years, AFC media enabled the industry's first 400 GB 3.5-inch drive, the Deskstar 7K400.

AFC media isn't the dominant way forward, however, and the immediate future lies in PMR.

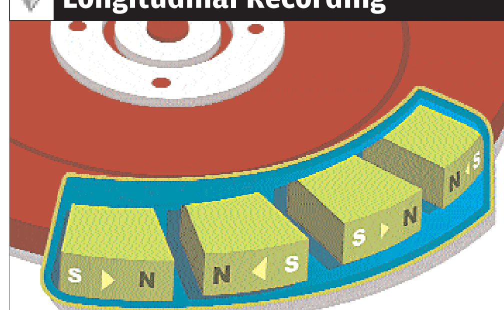
Perpendicular Magnetic Recording (PMR)

The highest areal density with LR has just recently crossed 100 gigabits psi. And researchers now believe that 120 gigabits psi will be 'too much', and that at that point, PMR will become dominant.

So what is PMR? Simple. The clusters that hold bits of data are not magnetised along the plane of the platter, but rather, 'vertically', that is, perpendicular to the plane of the platter. Think of it as high-rise buildings or skyscrapers in the place of sprawling condominiums. Or, looking at a platter held on the top of your palm, longitudinal recording creates horizontal bar magnets 'on' the platter; whereas PMR creates 'vertical' bar magnets.

Coercivity (the amount of magnetic field required to write data) is enhanced in PMR—meaning the data will be more stable against superparamagnetism. And the geometry lends

Longitudinal Recording



In longitudinal recording, the magnetisation happens in the plane of the platter. Think of the bits of data as little bar magnets lying along the platter